

Verification of Probing Security for Masked Circuits

A Constructive Approach and Theoretical Study

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5. **Result:** soundness proofs, gadgets verified that a comparable tool false-flags, and competitive performance.

Motivation & Model

Side-channel attacks

A cipher can be secure on paper and insecure on the bench.

- A real device leaks through power draw, timing, electromagnetic emanation; all of which vary with the *data* it handles.
- Differential power analysis (Kocher et al. 1999) correlates many power traces against a guess of an intermediate value, and may recover the key.
- In practice: DPA recovered keys from deployed smartcards; timing channels underlie Spectre and Meltdown.

So, the insecurity lies not in the mathematics behind the cipher, but in the way the hardware computes it.

The setting

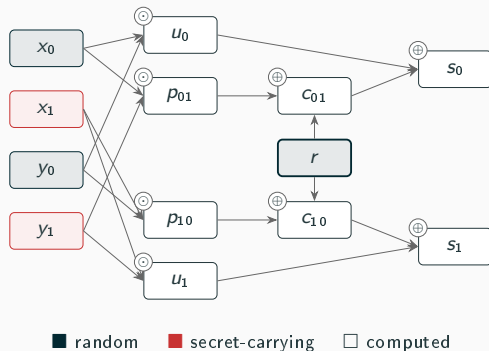
We study **arithmetic circuits** over the two-element field $\mathbb{F}_2$, built from two-input $\oplus$ (XOR) and $\odot$ (AND) gates.

Inputs are partitioned into three classes

- **secrets** $\mathcal{S}$ — the key material we protect,
- **randoms** $\mathcal{R}$ — fresh uniform bits (the masks), drawn together as a *tape* $\rho \in \{0, 1\}^{\mathcal{R}}$,
- **publics** $\mathcal{P}$ — known to everyone, held fixed.

Masking — and our running example

Boolean masking (Chari et al. 1999): split each secret into shares, $x = x_0 \oplus x_1$ with x_0 uniformly random, and compute only on shares, never on secrets.



DOM-AND (Groß et al. 2016): a masked AND gate, two shares per input, one fresh mask r .

- outputs re-share $s_0 \oplus s_1 = x \odot y$,
- cross terms p_{01}, p_{10} mix shares of *both* secrets,
- r re-masks them before the outputs.

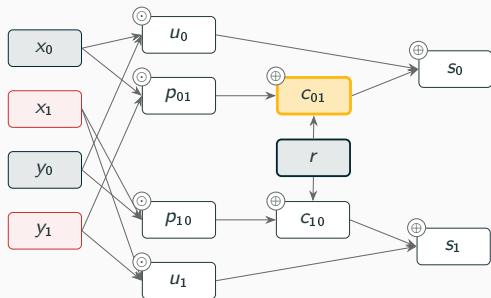
The probing model

Reasoning about noisy traces directly is unwieldy. Luckily we have an abstraction (Ishai et al. 2003):

d -probing security

An adversary picks any d wires $O = \{w_1, \dots, w_d\}$ and is handed their *exact* values. The circuit is d -probing secure when, for every such choice and every two secret assignments a, b , the observed values are identically distributed over the tape ρ :

$$\mathcal{L}_a(w_1, \dots, w_d) = \mathcal{L}_b(w_1, \dots, w_d).$$



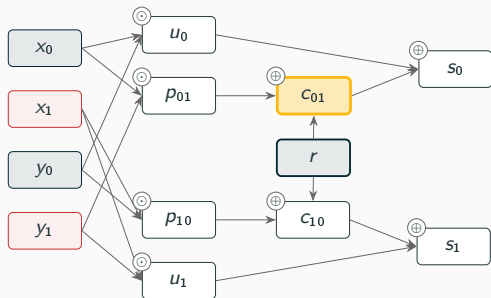
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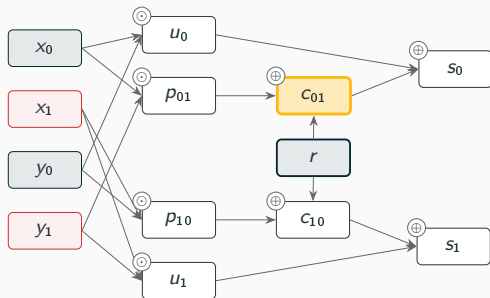
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- Security becomes a *combinatorial* property, decidable in principle.
- The abstraction is faithful: probing security implies noisy-leakage security (Duc et al. 2014).

The problem

Exact verification of d -probing security

Decide, for every one of the $\sum_{i \leq d} \binom{|\mathcal{W}|}{i}$ observation sets O , whether it is *independent of the secrets*:

$$\mathcal{L}_a(O) = \mathcal{L}_b(O) \quad \text{for all } a, b \in \{0, 1\}^{\mathcal{S}}.$$

- Secure $\Rightarrow$ every observation set accounted for,
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- Secure $\Rightarrow$ every observation set accounted for,
- Insecure $\Rightarrow$ a specific leaking observation as witness.
- Two difficulties arise: a *single* probe already asks a question about a distribution over all of $\{0, 1\}^{\mathcal{R}}$,
- and there are combinatorially many probes to settle.

How Hard Is One Probe?

Warm-up: three views of secret-independence

Fix a probe O (a tuple w of wires). Write $\mathcal{L}_a(O)$ for its distribution over a uniform tape $\rho \in \{0,1\}^{\mathcal{R}}$, and $N_a(o) = |\{\rho : O(\rho; a) = o\}|$ for the tape counts.

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3. **Coupling.** there exists a bijection Φ of the tapes such that $O(\rho; a) = O(\Phi(\rho); b)$.
 $\rightarrow$ *next section*

Theorem (thesis, Ch. 4)

Computing one tape count $N_a(o)$ is $\#P$ -complete (Valiant 1979). And deciding $\mathcal{L}_a(O) = \mathcal{L}_b(O)$ is $C=P$ -complete (Wagner 1986).

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- Hardness: reduction from EXACTSAT — one secret bit and one probed wire already encode it. (construction @ the appendix slides)
- So the hardness is intrinsic; no feature of masking causes it.

What the placement tells

- Deciding never costs more than counting ($C=P \subseteq P^{\#P}$), and no polynomial exact decision exists unless $P = NP$.
- Both classes sit *above the polynomial hierarchy* (due to Toda). A single probe is already there, and full verification universally quantifies over exponentially many such instances.

Consequence

We do not compute counts. We certify *constructively*: exhibit a relabelling of the randomness under which the secrets visibly cancel — a witness without counting, at the price of completeness!

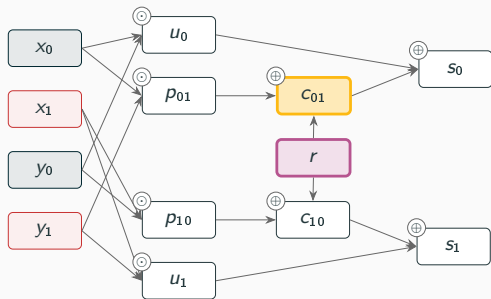
Security Is a Coupling

One substitution, on the running example

Probe the masked cross term: $c_{01} = x_0 \odot y_1 \oplus r$.

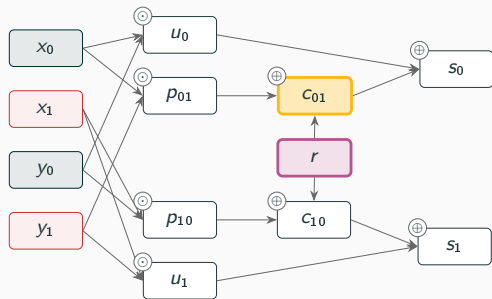
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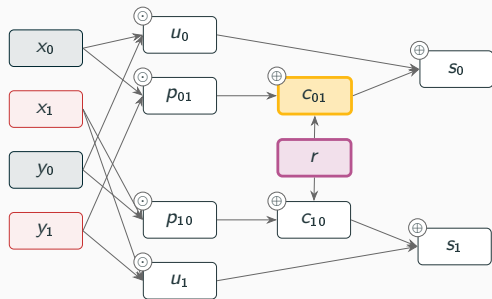
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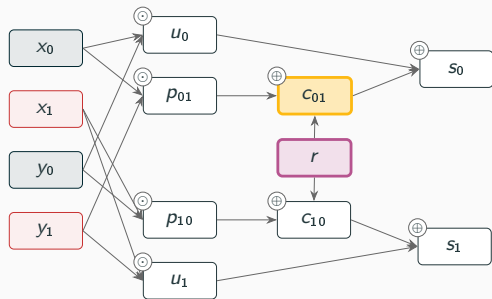
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- And σ induces a *bijection* of the tape space, translating the coordinate r — a *shear*.
- The context $x_0 \odot y_1$ mentions a share, so the *one* syntactic step induces a *family* of shears — fixing the secrets a picks $\hat{\sigma}^a$.

Key identity: $(w\sigma)(\rho; a) = w(\hat{\sigma}^a(\rho); a)$ — syntax and semantics agree, for every a .

To prove two distributions equal *without computing them*, link their sources ((Barthe et al. 2017) for probabilistic programs). Our two laws arise from the *same* uniform tape, so we link the tapes.

Definition (coupling function)

A *coupling function* from a to b for a probe O is a bijection $\Phi : \{0, 1\}^{\mathcal{R}} \rightarrow \{0, 1\}^{\mathcal{R}}$ with

$$O(\rho; a) = O(\Phi(\rho); b) \quad \text{for every tape } \rho.$$

- Bijectivity preserves the uniform law; the identity says a run under a is reproduced, observation for observation, under b . Hence we have equal laws.

Theorem

A probe O is secret-independent *iff* for every ordered pair a, b there is a coupling function from a to b for O .

Proof sketch.

- ($\Leftarrow$) A bijection merely re-indexes the tape space, so the counts behind the two laws agree.
- ($\Rightarrow$) Equal laws mean the outcome-fibres $\{\rho : O(\rho; a) = o\}$ come in matching sizes; match them block by block and glue. $\square$

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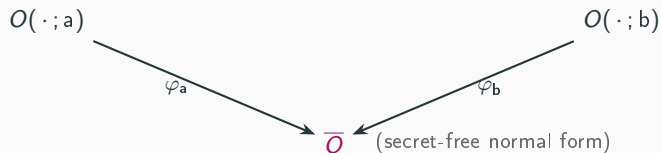
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This is an existence statement and not a procedure

Matching the fibres needs the counts $N_a(o)$, which are $\#P$ -hard. The theorem guarantees a coupling; it does not find one.

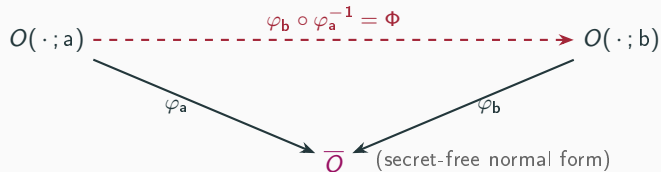
From substitutions to couplings

A derivation $\sigma_1, \dots, \sigma_K$ ending secret-free is a *recipe*: run under secrets a , it becomes the tape bijection $\varphi_a = \widehat{\sigma}_1^a \circ \dots \circ \widehat{\sigma}_K^a$. Applied syntactically, it yields $\overline{O} = O\sigma_1 \dots \sigma_K$.



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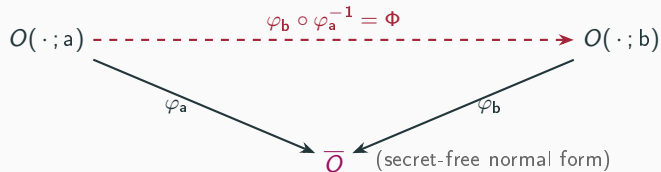
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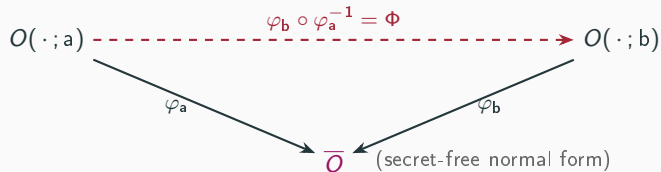
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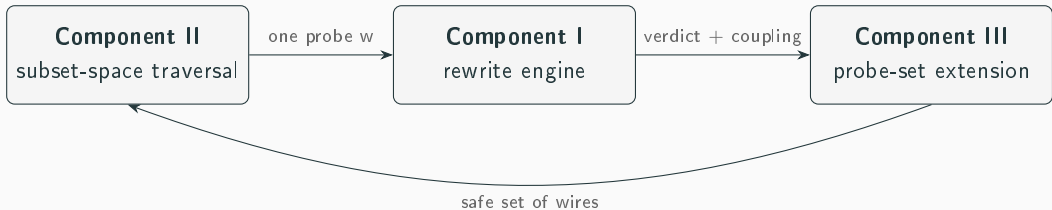
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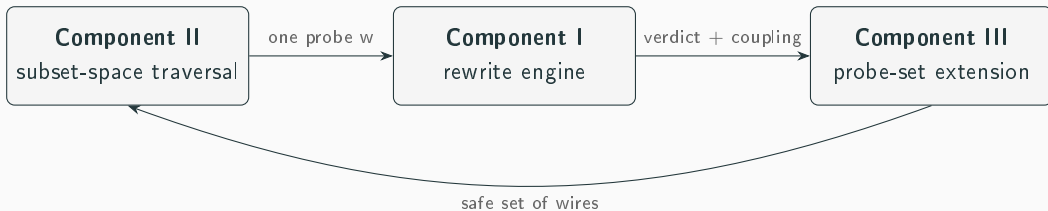
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- So a derivation is a **reusable certificate**: the design principle of Coppula.

Coppula: Three Components

Pipeline overview

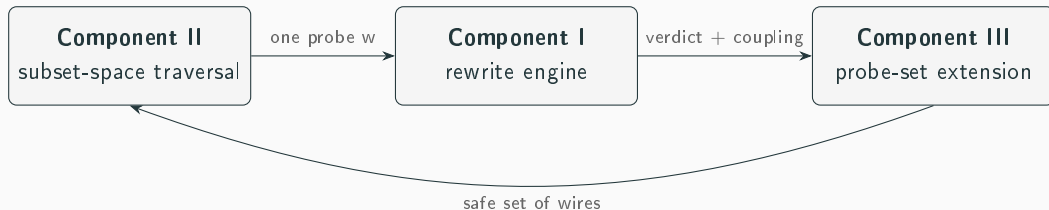


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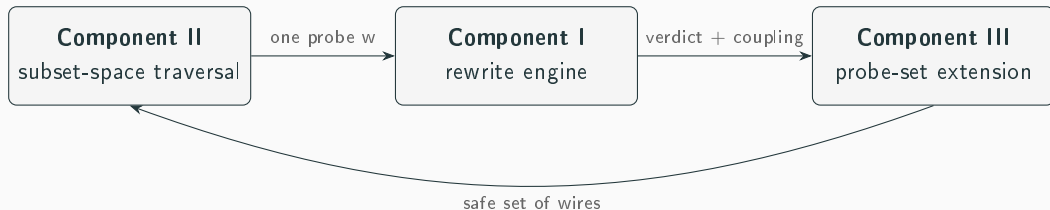
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- III amplifies one certificate into a whole *safe set*, shrinking the space II must sweep.

Component I: the rewrite engine

Decides one probe w by substituting randoms away, one at a time, driving the observation toward a secret-free form.

- **Tiered.** A cheap single-occurrence rule first (r occurs once across w : substitute, done); with a complete substitution loop behind it.

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On the example: $c_{01} \xrightarrow{r \mapsto x_0 \odot y_1 \oplus r} r$ — one step, Secure, coupling recorded.

Component I: why factoring

Some masks are buried inside *products*, where substitution cannot reach them.

- Substituting $r \mapsto e \oplus r$ needs r to occur as a *summand*; in $u \odot r \oplus u \odot v$ every occurrence of r is trapped inside a product.

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- Applied lazily between substitution attempts, until the probe discharges or no rule applies.

Component II: the representation

Certifying each d -subset in turn is infeasible. So the traversal never lists probes; it holds compact descriptions of whole families, in the manner of maskVerif (Barthe et al. 2019).

Factors and worklists

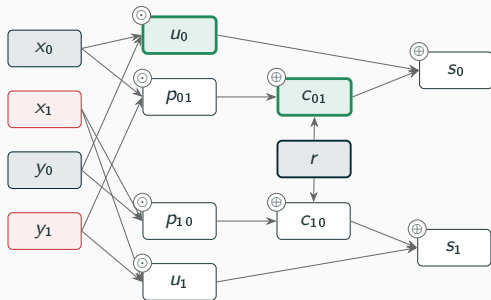
A *factor* (c, W) stands for “choose c wires from the set W ”. A *worklist*

$\langle (c_1, W_1), \dots, (c_m, W_m) \rangle$, the W_j disjoint, denotes all probes drawing c_j wires from each W_j : it holds $\prod_j \binom{|W_j|}{c_j}$ probes in a few words.

On DOM-AND (*order 2, for exposition*): the probe targets are the eight computed wires $\mathcal{W} = \{u_0, u_1, p_{01}, p_{10}, c_{01}, c_{10}, s_0, s_1\}$, and the whole verification task is the single factor

$$\langle (2, \mathcal{W}) \rangle, \quad \text{denoting all } \binom{8}{2} = 28 \text{ pairs.}$$

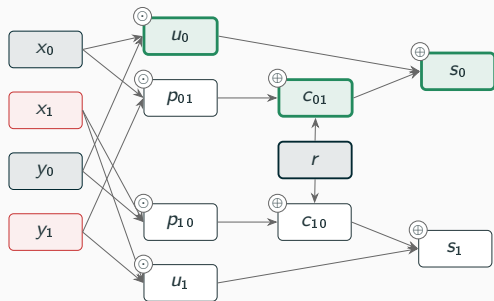
Component II: one recursion step, on our example



the illustrated probe set, in green

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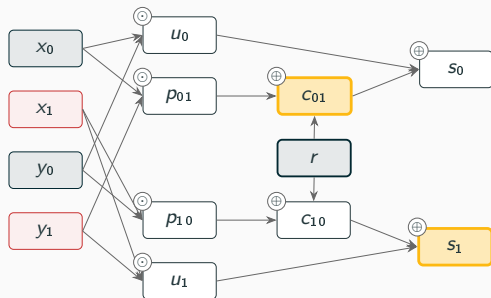
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the illustrated safe set, in green

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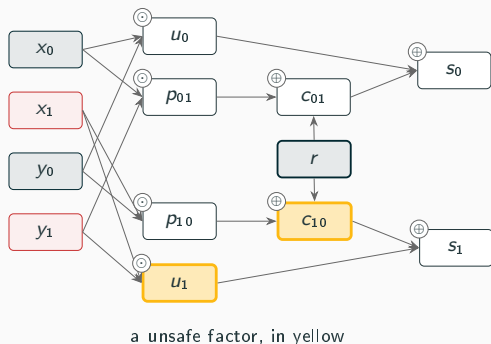
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a cross-term factor, in yellow

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 $28 = \binom{3}{2} + 3 \cdot 5 + \binom{5}{2} = 3 + 15 + 10$ (all-safe, mixed, all-unsafe). *Skip* the all-safe cell and recurse on the 25 that remain.

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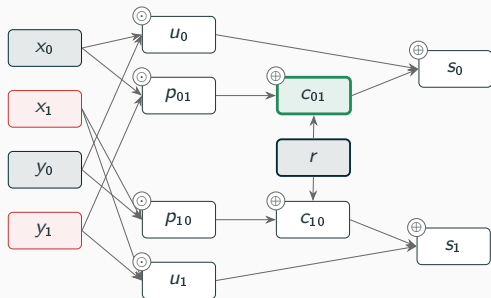


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Theorem (exhaustiveness, Ch. 5). Read set-wise, Vandermonde's identity says the cells *partition* the probes: every d -subset is visited exactly once.

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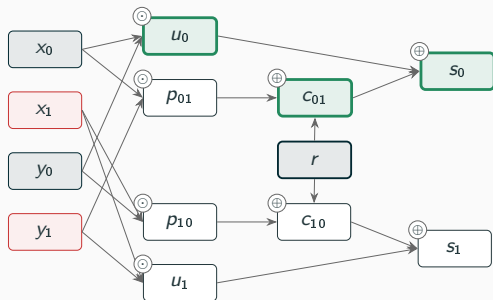
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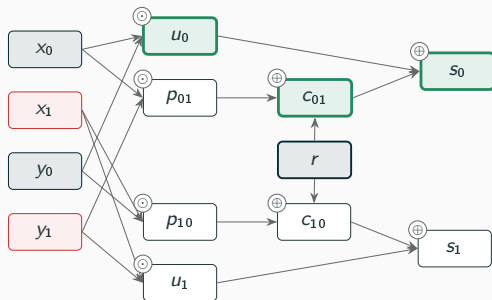
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- Here $\varphi = (r \mapsto r \oplus x_0 \odot y_1)$ also blinds u_0 and s_0 , as $u_0 \circ \varphi = x_0 \odot y_0$ and $s_0 \circ \varphi = x_0 \odot y_0 \oplus r$.

Component III: growing safe sets

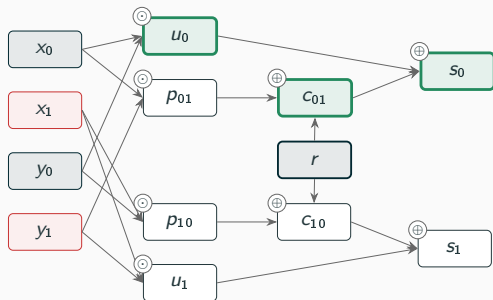
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- Exact test via *ANF*: a function depends on a variable iff it occurs in its normal form.
- One discharge settles the whole safe set; Component II needs less traversal.

Component III: three mechanisms

- **Closure** — build-time, entropy-style deduction ($H(u \mid w) = 0$). Local deduction on a scattered pool: cheap but weak.

Proposition (nesting, Ch. 5)

closure $\subseteq$ replay $\subseteq$ coupling — every wire the cheaper mechanisms admit, the coupling admits too. Coppula adopts the coupling approach.

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- **Replay** — re-run the derivation on the candidate wire (Barthe et al. 2015). An imperfect fit for us because we *factor*, and factoring is not a replayable step.
- **Coupling** — apply the recorded tape bijection and test the ANF endpoint. Exact, and independent of how the derivation was found.

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Component III: the test

Given the coupling φ and a candidate wire u from the pool: is $u \circ \varphi$ secret-free?

Phase 1 — fast reject (probabilistic, may only *reject*)

Sample tapes; evaluate $u \circ \varphi$ twice per tape, secrets drawn at random vs. all zero. A disagreement proves $u \circ \varphi$ moves with the secrets: drop u . Agreement proves nothing.

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Why blinded wires are safe to add. For admitted u , $u(\varphi_a(\rho); a) = \bar{u}(\rho)$ depends on ρ alone; as φ_a preserves the uniform law, the *joint* law of the safe set is the same under every secret (Theorem, Ch. 5).

Implementation & Evaluation

Coppula is implemented in **Lean 4**.

- **Hash-consed DAG** expression representation with canonicalization — structural sharing makes equality checks and substitutions cheap.
- **Multiplicative factoring** as a normal-form pass over the DAG, applied lazily.
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Design and proofs are implementation-agnostic; these choices simply make algorithms fast.

Evaluation: outcomes

Forty configurations at orders 1–4: ISW multiplications, DOM and TI ANDs, refresh gadgets, Keccak χ (DOM and TI sharings), and the TI S-boxes and quadratic sharings of Bilgin et al. Every verdict matches the ground truth, and every Insecure comes with a concrete witness observation.

- **DOM-AND** (2 shares): Secure at order 1, the witness pair $\{c_{01}, c_{10}\}$ at order 2 — our running example.
- **Four-share $x \oplus yz$ TI** (Bilgin et al. 2015): certified secure at orders 1 and 2, insecure at order 3 — the *expected* behaviour of a first-order threshold implementation.

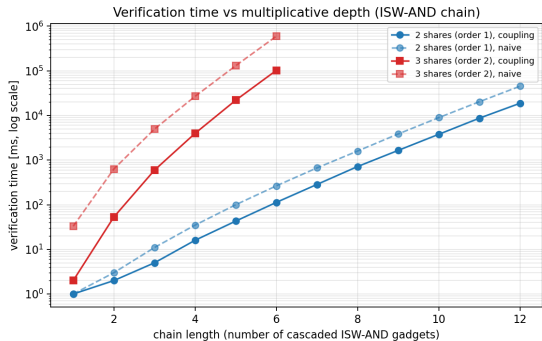
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- **The factoring family** ($d+1$ quadratic S-boxes, Q^3/Q^4): randomness-free sharings whose certification must factor first. A factoring-free, maskVerif-style engine false-reports Insecure on five; Coppula verifies every one.

(Full extension-comparison table @ the appendix slides.)

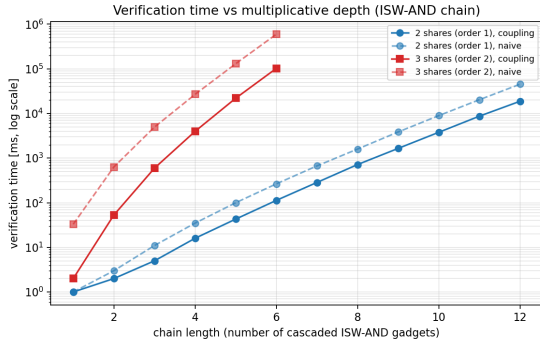
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Verification time versus circuit depth.

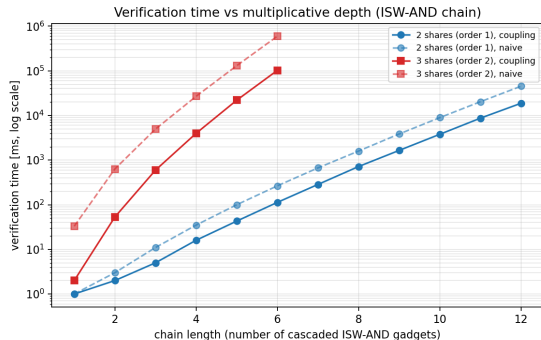
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- Closures keep the verdicts but lose the yield: 14 vs. 537 free wires on Fides, 632,834 vs. 4735 discharges on five-share ISW — the split degrades toward enumeration.
- The extension pays for itself: Fides certifies 537 wires from 55 discharges; five-share ISW falls from ~ 609 s (no extension) to under 2 s.

Wrap-up

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- **Composing couplings.** for unions of safe sets using couplings as a guide.
- **More.** smarter factoring heuristics; exploiting circuit symmetries to quotient the subset space.

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3. **Coppula.** Made up of a sound rewrite engine (with factoring), an exhaustive subset-space traversal, and a probe-set extension with a proven mechanism hierarchy closure $\subseteq$ replay $\subseteq$ coupling.
4. **Evaluation.** We present exact verdicts, with witnesses, across a suite of masked gadgets, and the coupling extension wins in coverage.

Thank you!

Questions?

Backup

Backup: extension comparison (I/IV)

Each entry: Coupling / Replay / Closure. Verdict S = Secure, I = Insecure; † marks replay's false-Insecures.

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
ISW multiplication					
2 sh.	1	S/S/S	0/0/1	7/7/15	2/2/0
3 sh.	2	S/S/S	3/1/34	34/34/309	31/31/8
4 sh.	3	S/S/S	46/37/3190	338/338/10770	437/437/439
5 sh.	4	S/S/S	1125/845/409472	4735/4735/632834	8078/8078/37513
DOM AND					
2 sh.	1	S/S/S	1/0/0	11/11/15	2/2/0
2 sh.	2	I/I/I	0/0/1	2/2/6	0/0/2
3 sh.	2	S/S/S	5/4/35	88/88/325	44/44/7
4 sh.	3	S/S/S	104/90/3419	859/859/13291	784/784/532

Backup: extension comparison (II/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
Trichina AND					
well-formed	1	S/S/S	0/1/1	11/11/15	2/2/0
mis-associated	1	I/I/I	1/0/1	10/10/12	1/1/0
TI AND					
3 sh.	1	S/S/S	0/0/1	9/9/27	9/9/0
3 sh.	2	I/I/I	1/1/1	10/10/10	5/5/2
Mask refreshing					
additive, 3 sh.	2	S/S/S	0/0/0	6/6/8	1/1/0
additive, 4 sh.	3	S/S/S	1/0/1	11/11/17	6/6/0
ISW, 3 sh.	2	S/S/S	0/0/0	6/6/10	4/4/0
ISW, 4 sh.	3	S/S/S	0/1/2	12/12/51	24/24/0
Keccak χ					
DOM, 2 sh.	1	S/S/S	14/11/29	49/49/107	29/29/5
DOM, 3 sh.	2	S/S/S	2807/1797/11563	2865/2865/14047	2671/2671/645
TI, 3 sh.	1	S/S/S	21/14/103	43/43/183	68/68/5
TI, 3 sh.	2	I/I/I	21/14/72	57/57/352	92/92/27

Backup: extension comparison (III/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
TI S-boxes and quadratic sharings					
Q_{12}^4 direct, 2 sh.	1	S/I [†] /S	2/2/5	27/26/47	16/16/10
Q_{12}^4 TI, 3 sh.	1	S/S/S	5/4/25	25/25/83	30/30/6
Fides AB1, 4 sh.	1	S/S/S	1941/972/36461	55/55/1127	537/537/14
$x + yz$ direct, 3 sh.	1	S/S/S	1/0/3	7/7/29	10/10/0
$x + yz$ w/ CT, 3 sh.	1	S/S/S	1/1/3	11/11/29	9/9/0
$x + yz$, 4 sh.	2	S/S/S	11/8/170	78/78/767	115/115/20
$x + yz$, 4 sh.	3	I/I/I	11/7/8	102/102/72	99/99/15
$x + yz + xyz$, 4 sh.	1	S/S/S	45/18/175	31/31/167	63/63/1
xy virtual var.	1	S/S/S	3/1/17	9/9/55	23/23/0
xy virtual share	1	S/S/S	1/0/7	9/9/37	15/15/0
xy , $4 \rightarrow 3$ sh.	1	S/S/S	1/1/3	11/11/25	8/8/0

Backup: extension comparison (IV/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
$d+1$ quadratic S-box family					
Q_1^3	1	S/S/S	0/1/0	13/13/23	11/11/10
Q_2^3	1	S/I ⁺ /S	2/1/4	25/24/43	13/13/6
Q_3^3	1	I/I/I	4/2/7	18/18/32	22/22/2
Q_4^4	1	S/S/S	1/0/1	15/15/27	14/14/14
Q_{12}^4	1	S/I ⁺ /S	2/2/5	25/24/47	15/15/8
Q_{293}^4	1	S/I ⁺ /S	4/2/6	31/24/59	20/20/8
Q_{294}^4	1	S/S/S	1/1/3	23/23/39	16/16/12
Q_{299}^4	1	S/I ⁺ /S	5/2/13	35/18/75	26/22/8
Q_{300}^4	1	I/I/I	4/3/8	18/18/32	24/24/2

The two genuinely insecure classes, Q_3^3 and Q_{300}^4 , are exactly the quadratic classes that admit no uniform three-share TI (Bilgin et al. 2015).

Backup: C=P membership, sketch

Encode disagreement of the two laws as a single gap function:

$$f(C, O, a, b) = \sum_o (N_a(o) - N_b(o))^2.$$

Each N is a #P function; GapP is closed under subtraction and multiplication (Fenner et al. 1994); the sum has $2^k \leq 2^d$ terms, constant for fixed d . A sum of squares vanishes iff every term does, so

$$\mathcal{L}_a(O) = \mathcal{L}_b(O) \iff f = 0,$$

and the problem is the vanishing set of a gap function, which is exactly C=P.

Backup: the ExactSat reduction

Reduce from the equality form of EXACTSAT: given formulas φ_0, φ_1 over variables $x_1, \dots, x_m$, decide $\#\varphi_0 = \#\varphi_1$ (C=P-complete, (Wagner 1986)).

Build a circuit with one secret bit s , the x_i as randoms, and the single wire

$$w = \varphi_0 \oplus (s \odot (\varphi_0 \oplus \varphi_1)),$$

realising the formulas over $\oplus$ and $\odot$. Then w reads φ_0 when $s = 0$ and φ_1 when $s = 1$, so the one-bit probe $\{w\}$ is secret-independent *iff* $\#\varphi_0 = \#\varphi_1$.

The construction is polynomial, and one secret bit with one probed wire suffices: no feature of masking is what makes the problem hard.

Backup: security is a coupling, full sketch

($\Leftarrow$) Reindex by $\rho' = \varphi(\rho)$:

$$N_a(o) = |\{\rho : O(\rho; a) = o\}| = |\{\rho : O(\varphi(\rho); b) = o\}| = N_b(o).$$

($\Rightarrow$) Sort the tapes by outcome: $F_a(o) = \{\rho : O(\rho; a) = o\}$. As o varies these *fibres* partition $\{0, 1\}^{\mathcal{R}}$, and equal laws mean matching fibres have equal size, $|F_a(o)| = |F_b(o)|$. So pick, for each o , any bijection $F_a(o) \rightarrow F_b(o)$; their union is a bijection φ of the whole tape space, and it sends every ρ to a tape with the *same* observation, giving us a coupling function. $\square$

Backup: replay vs. factoring

The derivations of Barthe et al. (2015) replay two rule kinds: OPT (substitution) and CONV (ANF normalisation) — for them, ANF is a *replayed rule*.

Our derivations interleave substitutions with *multiplicative factoring*, and a factoring step has no tape-level counterpart to replay. Replaying the substitution steps alone on a new wire can spuriously fail, producing false-Insecure verdicts (observed in the evaluation, the † rows).

Hence the coupling mechanism: apply the recorded bijection $\hat{\sigma}_1 \circ \dots \circ \hat{\sigma}_k$ semantically and test the ANF endpoint — exact, and indifferent to how the derivation was found.

Backup: positioning

- **maskVerif** (Barthe et al. 2019): language-based, the traversal's main inspiration; replay-style extension; no factoring.
- **REBECCA / Fourier-analytic** (Bloem et al. 2018): correlation sets over Fourier coefficients; approximate in general.
- **SILVER** (Knichel et al. 2020): exact, BDD-based model counting; exhaustive per-probe distributions, different scaling profile.
- **Coppula**: exact like SILVER, constructive like maskVerif — and the certificate (the coupling) is itself an object with structure we exploit.

Backup: the four-share TI netlist

Four-share TI of $x \oplus (y \odot z)$ (Bilgin et al. 2015); each input four-shared, sixteen cross products $y_i \odot z_j$:

$$\begin{aligned} F_1 &= x_2 \oplus \bigoplus_{i,j \in \{2,3,4\}} y_i z_j, & F_2 &= x_3 \oplus y_1 z_3 \oplus y_1 z_4 \oplus y_3 z_1 \oplus y_4 z_1 \oplus y_1 z_1, \\ F_3 &= x_4 \oplus y_1 z_2 \oplus y_2 z_1, & F_4 &= x_1. \end{aligned}$$

A *first-order* threshold implementation — never claimed beyond order one. Coppula certifies orders 1 and 2 and finds the order-3 witness: a counterexample on a gadget with a *known* failure order, not a flaw.

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-  Knichel, David, Pascal Sasdrich, and Amir Moradi (2020). **“SILVER – Statistical Independence and Leakage Verification”**. In: *Advances in Cryptology – ASIACRYPT 2020*. Vol. 12491. Lecture Notes in Computer Science. Springer, pp. 787–816. DOI: 10.1007/978-3-030-64837-4_26.
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